

Unless otherwise stated, each problem is worth 4 points. Show all of your work.

(a) Find the inverse on the specified interval express it in the form $y = f^{-1}(x)$. (b) Verify that $f^{-1}(f(x)) = x$. (c) State the range and domain of $f^{-1}(x)$.

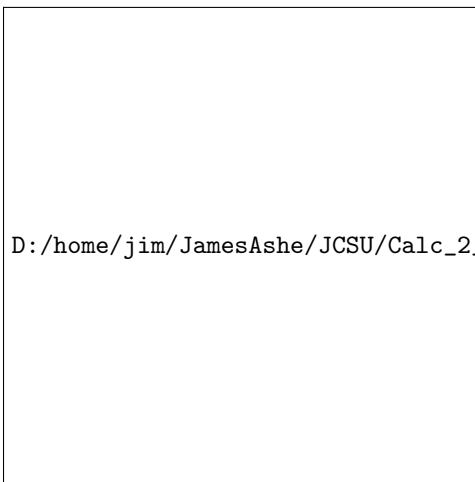
1. $f(x) = \sqrt{x-3} + 1$ for $x \geq 3$ [6 pts]

Find the derivative of the inverse of the following functions at the specified point on the graph of the inverse function without finding f^{-1} .

2. $f(x) = x^4 - 2x^3 + 1$, at $(0, 1)$ (point on f^{-1}).

Use the graph of $f(x)$ to sketch $f^{-1}(x)$.

3. [6 pts] (The dashed line is $y = x$.)



D:/home/jim/JamesAshe/JCSU/Calc_2_MTH_232/images/graph_the_inverse.png

Perform the indicated operation.

4. $\frac{d}{dx} (\ln x + 1)$

5. $\frac{d}{dx} 3e^x \ln x$

6. $\frac{d}{dx} (\ln(\cos x) - 2x)$

7. $\frac{d}{dx} e^{-2x+x^2}$

8. $\frac{d}{dt} \frac{e^t}{e^t + 1}$

9. $\int \left(3e^{-3x} + \frac{1}{x} \right) dx$

10. $\frac{d}{dx} 5 \cdot 4^x dx$

11. $\int_{-1}^1 9^x [6 \text{ pts}]$

12. $\int x e^{-4x} dx$

13. $\int x \cos 4x dx$

14. $\int \frac{4 \ln x}{x^5} dx$

15. $\int_0^{\ln 1} 3x e^x dx$ [6 pts]